# ECE238L Computer Logic Design

# Lab 3: Structural Design of

# Multiplexers and Decoders

Section 003: Wednesday 2:00-4:45

Name: Tomas Salaz

# Gates VHDL Source Code, Simulation Code and Constraints

Gates.vhd

C:/Users/tsalaz25/ECE238L/Gates/Gates.srcs/sources_1/new/Gates.vhd

```vhdl
-- Create Date: 02/25/2025 12:58:49 PM
-- Design Name:
-- Module Name: Gates - Behavioral
-- Project Name:
-- Target Devices:
-- Tool Versions:
-- Description:
--
-- Dependencies:
--
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:
--
----------------------------------------------------------------------------------


library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity Gates is
    Port ( A, B : in STD_LOGIC;
           G_out : out STD_LOGIC_VECTOR (7 downto 0));
end Gates;

architecture Behavioral of Gates is

begin
  G_out(0) <= (A); --Buffer
  G_out(1) <= not A;
  G_out(2) <= A and B;
  G_out(3) <= A or B;
  G_out(4) <= A nand B;
  G_out(5) <= A nor B;
  G_out(6) <= A xor B;
  G_out(7) <= A xnor B;
end Behavioral;
```

Gates_tb.vhd

C:/Users/tsalaz25/ECE238L/Gates/Gates.srcs/sim_1/new/Gates_tb.vhd

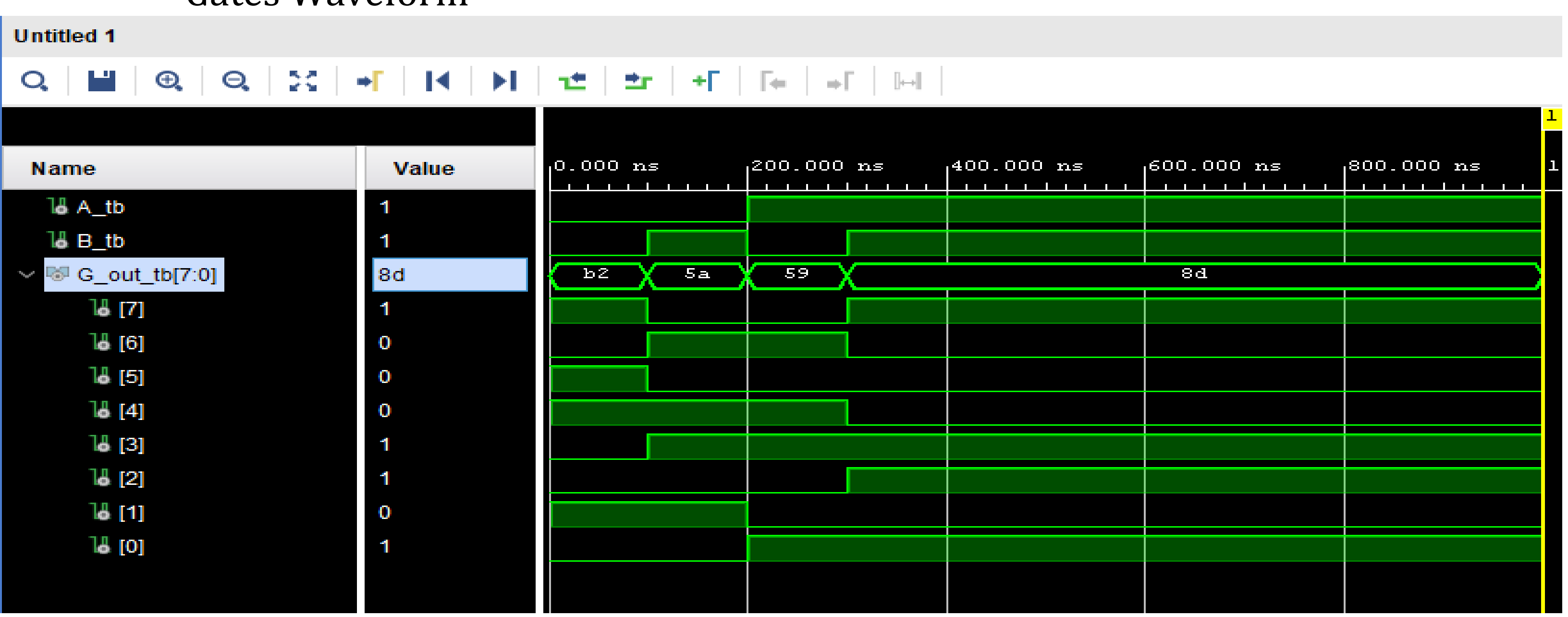
```vhdl
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:
--
----------------------------------------------------------------------------------


library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity Gates_tb is
--  Port ( );
end Gates_tb;

architecture bench of Gates_tb is

component Gates
  port (A, B : in STD_LOGIC;
        G_out: out STD_LOGIC_VECTOR(7 downto 0));
end component;

signal A_tb, B_tb : STD_LOGIC;
signal G_out_tb : STD_LOGIC_VECTOR(7 downto 0);

begin
uut : Gates port map (A => A_tb,
                      B => B_tb,
                      G_out => G_out_tb);
stimulus : process
begin
    A_tb <= '0' ; B_tb <= '0' ; wait for 100 ns;
    A_tb <= '0' ; B_tb <= '1' ; wait for 100 ns;
    A_tb <= '1' ; B_tb <= '0' ; wait for 100 ns;
    A_tb <= '1' ; B_tb <= '1' ; wait for 100 ns;
wait;
end process;
end bench;
```

Gates_constr.xdc

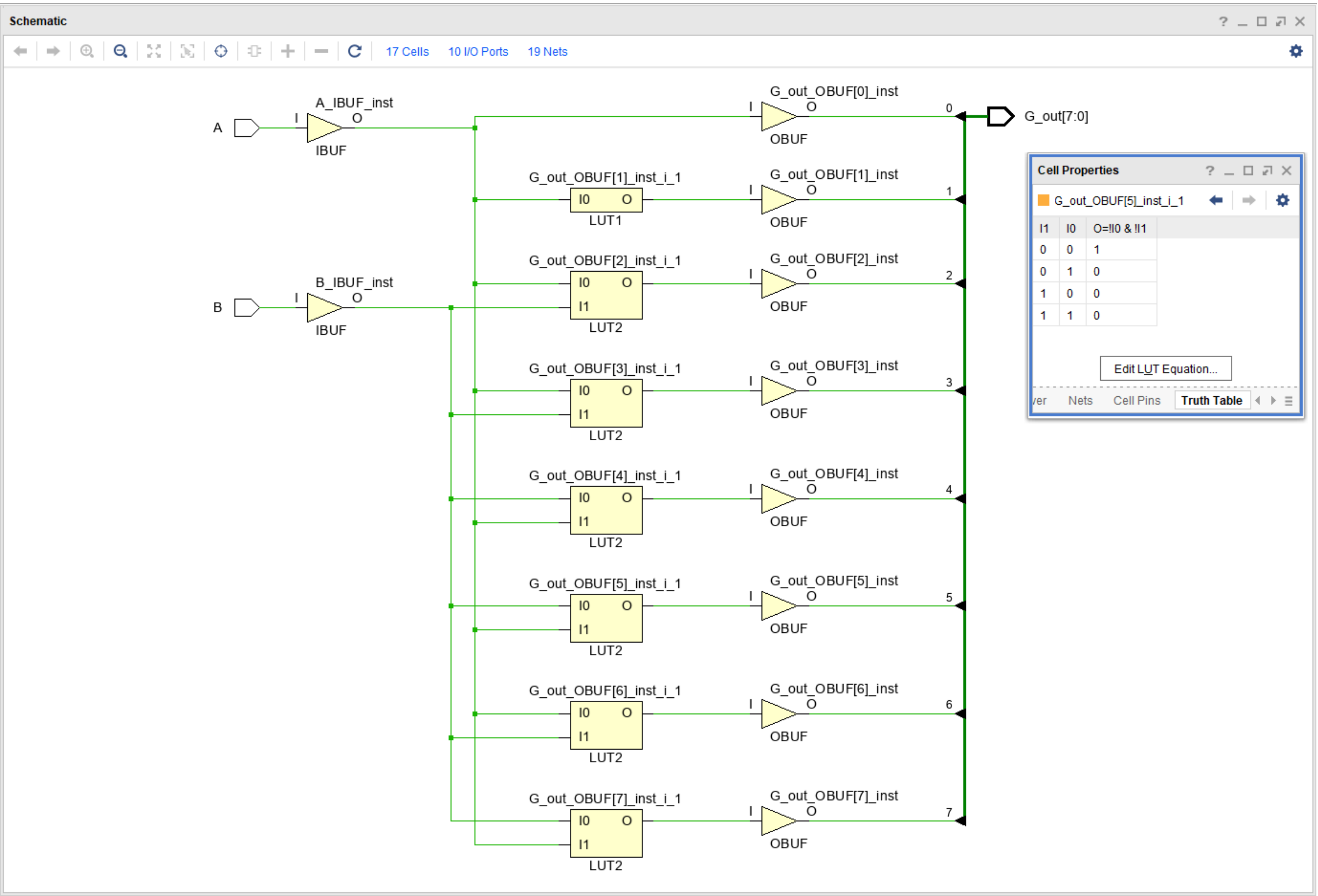
C:/Users/tsalaz25/ECE238L/Gates/Gates.srcs/constrs_1/new/Gates_constr.x

```
set_property IOSTANDARD LVCMOS33 [get_ports {G_out[7]}]
set_property IOSTANDARD LVCMOS33 [get_ports {G_out[6]}]
set_property IOSTANDARD LVCMOS33 [get_ports {G_out[5]}]
set_property IOSTANDARD LVCMOS33 [get_ports {G_out[4]}]
set_property IOSTANDARD LVCMOS33 [get_ports {G_out[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {G_out[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {G_out[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {G_out[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports A]
set_property IOSTANDARD LVCMOS33 [get_ports B]
set_property PACKAGE_PIN J15 [get_ports A]
set_property PACKAGE_PIN L16 [get_ports B]
set_property PACKAGE_PIN H17 [get_ports {G_out[0]}]
set_property PACKAGE_PIN K15 [get_ports {G_out[1]}]
set_property PACKAGE_PIN J13 [get_ports {G_out[2]}]
set_property PACKAGE_PIN N14 [get_ports {G_out[3]}]
set_property PACKAGE_PIN R18 [get_ports {G_out[4]}]
set_property PACKAGE_PIN V17 [get_ports {G_out[5]}]
set_property PACKAGE_PIN U16 [get_ports {G_out[6]}]
set_property PACKAGE_PIN V16 [get_ports {G_out[7]}]
```

# Gates Waveform

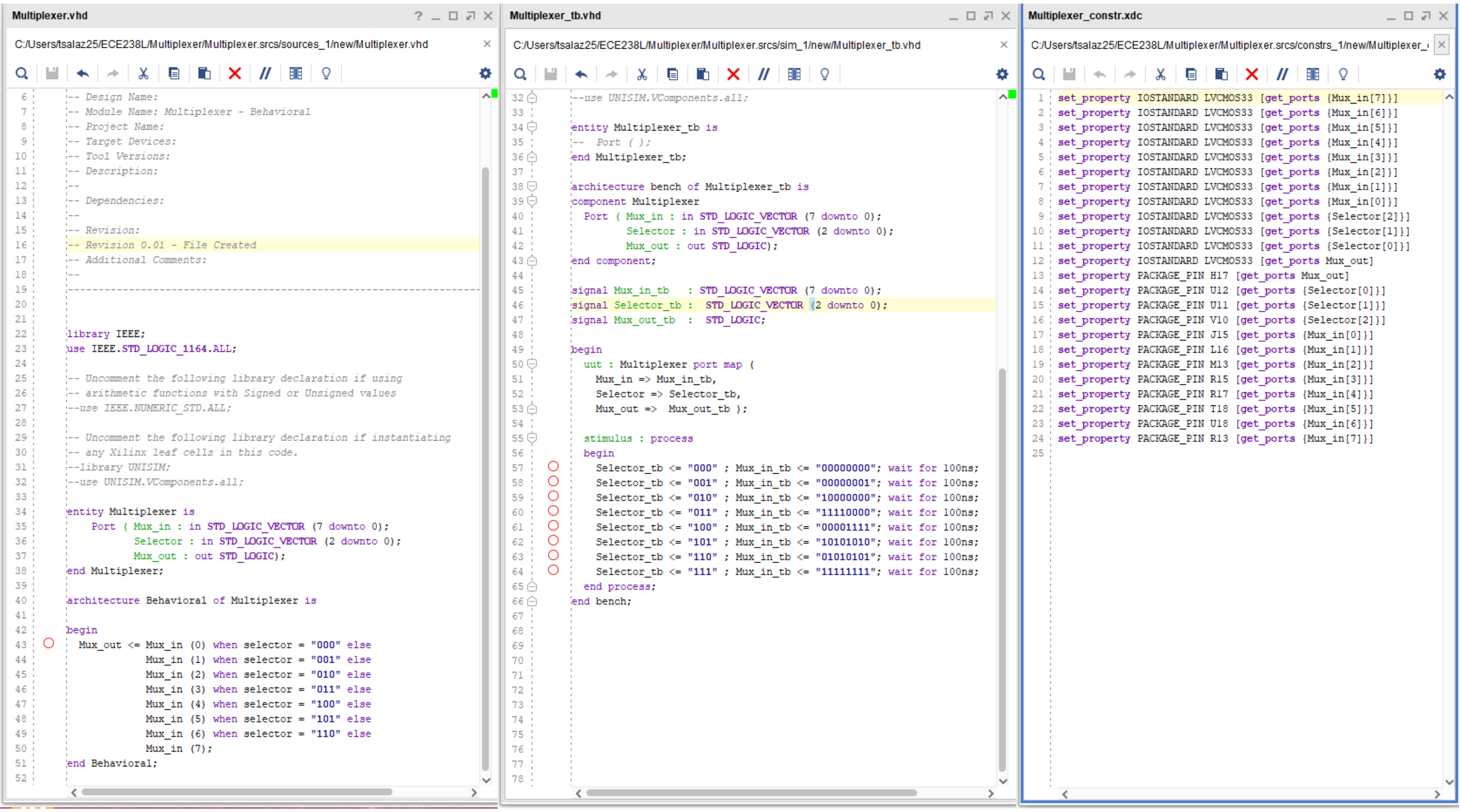


# Gates Truth Table and Schematic




| I1 | I0 | O=!I0 & !I1 |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

# Multiplexer VHDL Source Code, Simulation Code and Constraints



Multiplexer.vhd

C:/Users/tsalaz25/ECE238L/Multiplexer/Multiplexer.srcs/sources_1/new/Multiplexer.vhd

```vhdl
-- Design Name:
-- Module Name: Multiplexer - Behavioral
-- Project Name:
-- Target Devices:
-- Tool Versions:
-- Description:
--
-- Dependencies:
--
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:
--
----------------------------------------------------------------------------------


library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity Multiplexer is
    Port ( Mux_in : in STD_LOGIC_VECTOR (7 downto 0);
           Selector : in STD_LOGIC_VECTOR (2 downto 0);
           Mux_out : out STD_LOGIC);
end Multiplexer;

architecture Behavioral of Multiplexer is

begin
  Mux_out <= Mux_in (0) when selector = "000" else
             Mux_in (1) when selector = "001" else
             Mux_in (2) when selector = "010" else
             Mux_in (3) when selector = "011" else
             Mux_in (4) when selector = "100" else
             Mux_in (5) when selector = "101" else
             Mux_in (6) when selector = "110" else
             Mux_in (7);
end Behavioral;
```

Multiplexer_tb.vhd

C:/Users/tsalaz25/ECE238L/Multiplexer/Multiplexer.srcs/sim_1/new/Multiplexer_tb.vhd

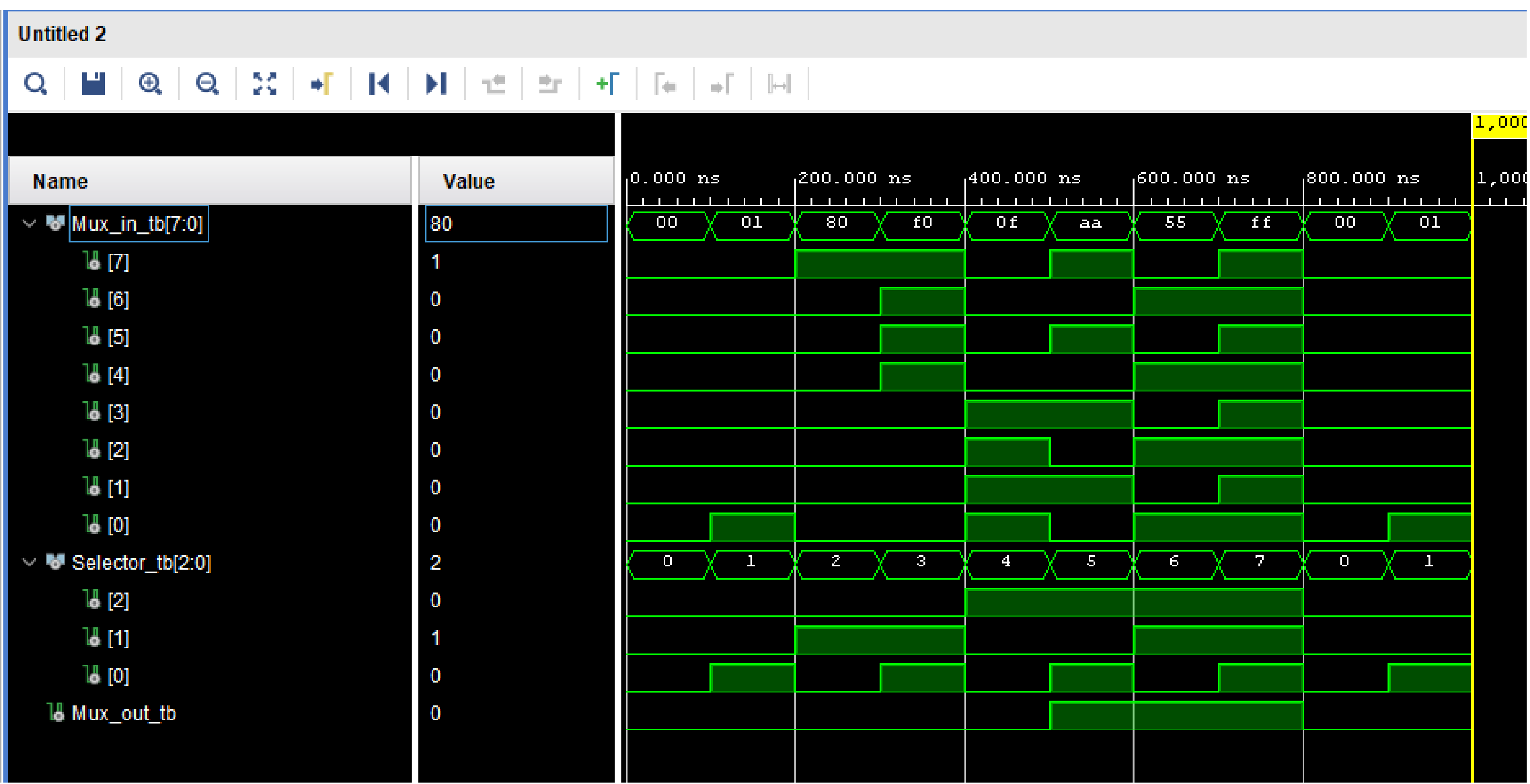
```vhdl
--use UNISIM.VComponents.all;

entity Multiplexer_tb is
--  Port ( );
end Multiplexer_tb;

architecture bench of Multiplexer_tb is
component Multiplexer
  Port ( Mux_in : in STD_LOGIC_VECTOR (7 downto 0);
         Selector : in STD_LOGIC_VECTOR (2 downto 0);
         Mux_out : out STD_LOGIC);
end component;

signal Mux_in_tb   : STD_LOGIC_VECTOR (7 downto 0);
signal Selector_tb :  STD_LOGIC_VECTOR (2 downto 0);
signal Mux_out_tb  :  STD_LOGIC;

begin
  uut : Multiplexer port map (
    Mux_in => Mux_in_tb,
    Selector => Selector_tb,
    Mux_out =>  Mux_out_tb );

  stimulus : process
  begin
    Selector_tb <= "000" ; Mux_in_tb <= "00000000"; wait for 100ns;
    Selector_tb <= "001" ; Mux_in_tb <= "00000001"; wait for 100ns;
    Selector_tb <= "010" ; Mux_in_tb <= "10000000"; wait for 100ns;
    Selector_tb <= "011" ; Mux_in_tb <= "11110000"; wait for 100ns;
    Selector_tb <= "100" ; Mux_in_tb <= "00001111"; wait for 100ns;
    Selector_tb <= "101" ; Mux_in_tb <= "10101010"; wait for 100ns;
    Selector_tb <= "110" ; Mux_in_tb <= "01010101"; wait for 100ns;
    Selector_tb <= "111" ; Mux_in_tb <= "11111111"; wait for 100ns;
  end process;
end bench;
```

Multiplexer_constr.xdc

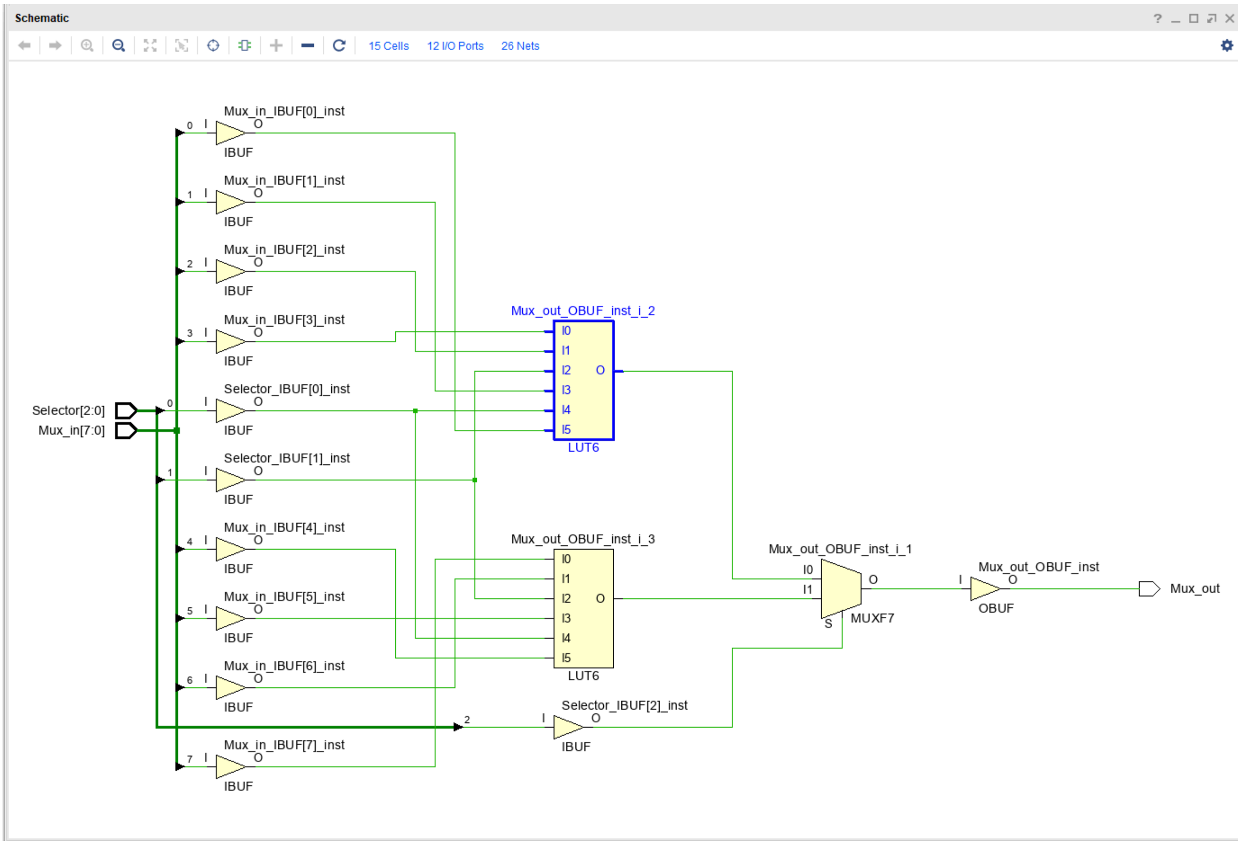
C:/Users/tsalaz25/ECE238L/Multiplexer/Multiplexer.srcs/constrs_1/new/Multiplexer_

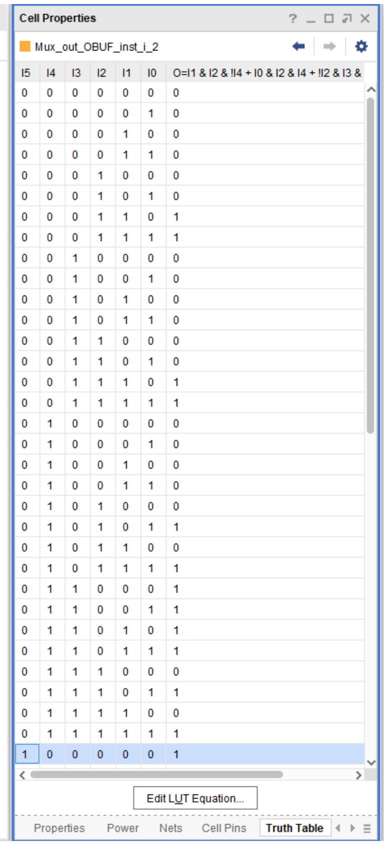
```
set_property IOSTANDARD LVCMOS33 [get_ports {Mux_in[7]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Mux_in[6]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Mux_in[5]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Mux_in[4]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Mux_in[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Mux_in[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Mux_in[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Mux_in[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Selector[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Selector[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Selector[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports Mux_out]
set_property PACKAGE_PIN H17 [get_ports Mux_out]
set_property PACKAGE_PIN U12 [get_ports {Selector[0]}]
set_property PACKAGE_PIN U11 [get_ports {Selector[1]}]
set_property PACKAGE_PIN V10 [get_ports {Selector[2]}]
set_property PACKAGE_PIN J15 [get_ports {Mux_in[0]}]
set_property PACKAGE_PIN L16 [get_ports {Mux_in[1]}]
set_property PACKAGE_PIN M13 [get_ports {Mux_in[2]}]
set_property PACKAGE_PIN R15 [get_ports {Mux_in[3]}]
set_property PACKAGE_PIN R17 [get_ports {Mux_in[4]}]
set_property PACKAGE_PIN T18 [get_ports {Mux_in[5]}]
set_property PACKAGE_PIN U18 [get_ports {Mux_in[6]}]
set_property PACKAGE_PIN R13 [get_ports {Mux_in[7]}]
```

# Multiplexer Waveform



# Multiplexer Truth Table and Schematic




| I5 | I4 | I3 | I2 | I1 | I0 | O=I1 & I2 & !I4 + I0 & I2 & I4 + !I2 & I3 & |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 | 1 | 0 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 0 | 1 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 | 1 | 0 | 1 |
| 0 | 0 | 1 | 1 | 1 | 1 | 1 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 | 1 | 0 | 0 |
| 0 | 1 | 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 | 1 | 0 | 0 |
| 0 | 1 | 0 | 1 | 1 | 1 | 1 |
| 0 | 1 | 1 | 0 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 | 1 | 1 | 1 |
| 0 | 1 | 1 | 1 | 0 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 | 1 |

# Display Decoder VHDL Source Code, Simulation Code and Constraints

DisplayDecoder.vhd

C:/Users/tsalaz25/ECE238L/DisplayDecoder/DisplayDecoder.srcs/sources_1/new/DisplayDec

```vhdl
-- Revision:
-- Revision 0.01 - File Created
-- Additional Comments:
--
----------------------------------------------------------------------------------


library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity DisplayDecoder is
    Port ( Disp_in : in STD_LOGIC;
           Cathode_7SD : out STD_LOGIC_VECTOR (7 downto 0);
           Anode_7SD : out STD_LOGIC_VECTOR (7 downto 0));
end DisplayDecoder;

architecture Behavioral of DisplayDecoder is

begin
  process (Disp_in)
  begin
  --If then to display 'L' = 1 or 'H' = 0
  --dependeing on Disp_in
   if Disp_in = '1'
     then Cathode_7SD <= "10010001";
   else
     Cathode_7SD <= "11100011";
   end if;
   end process;
   Anode_7SD <= "11111110";
end Behavioral;
```

DisplayDecoder_tb.vhd

C:/Users/tsalaz25/ECE238L/DisplayDecoder/DisplayDecoder.srcs/sim_1/new/DisplayDecode

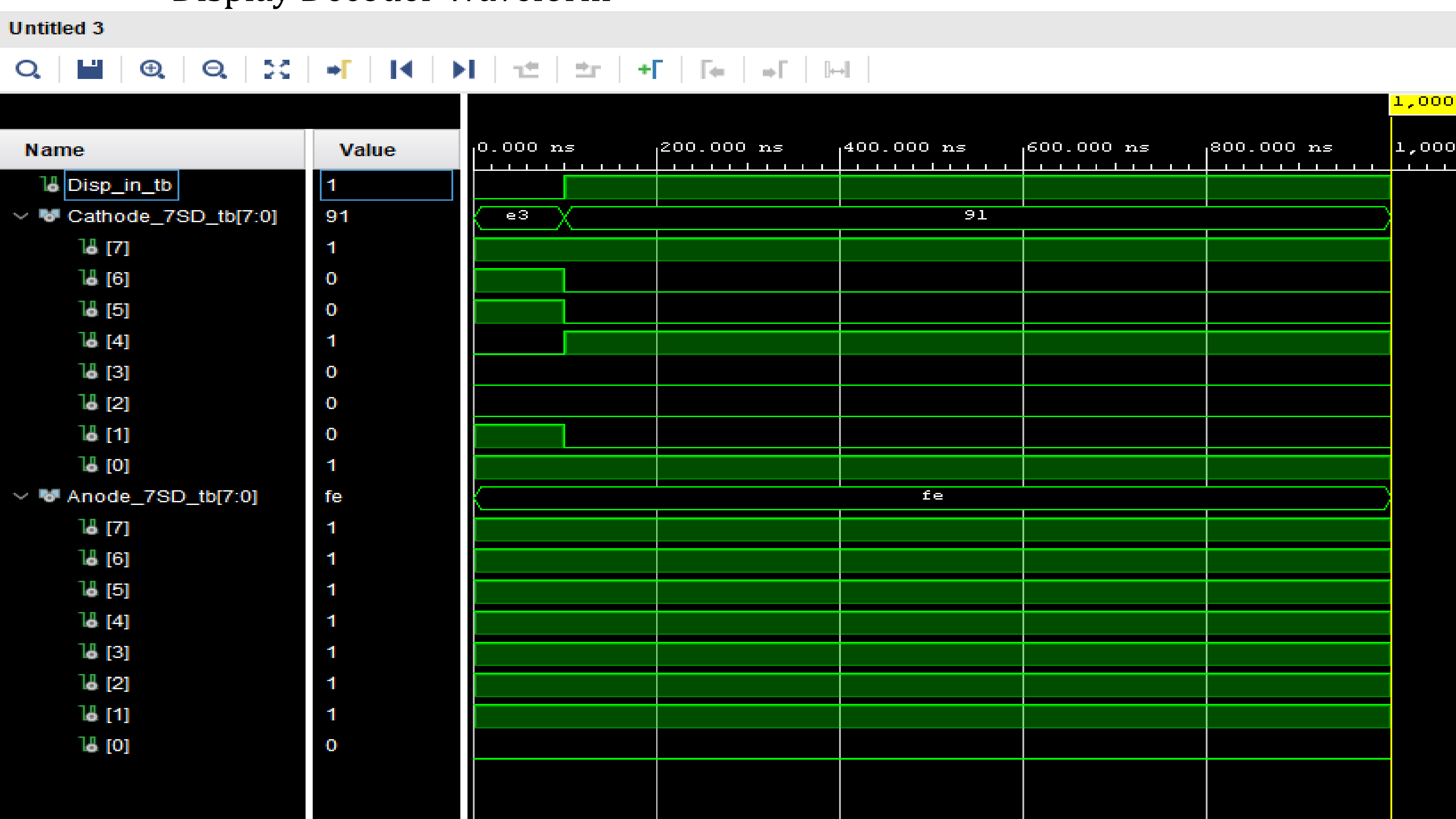
```vhdl


library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity DisplayDecoder_tb is
--  Port ( );
end DisplayDecoder_tb;

architecture bench of DisplayDecoder_tb is
component DisplayDecoder
  Port ( Disp_in : in STD_LOGIC;
         Cathode_7SD : out STD_LOGIC_VECTOR (7 downto 0);
         Anode_7SD : out STD_LOGIC_VECTOR (7 downto 0));
end component;

signal Disp_in_tb : STD_LOGIC;
signal Cathode_7SD_tb , Anode_7SD_tb : STD_LOGIC_VECTOR (7 downto 0)

begin
uut : DisplayDecoder port map (
  Disp_in => Disp_in_tb,
  Cathode_7SD => Cathode_7SD_tb,
  Anode_7SD => Anode_7SD_tb);

stimulus : process
  begin
    Disp_in_tb <= '0' ; wait for 100 ns;
    Disp_in_tb <= '1' ; wait for 100 ns;
  wait;
  end process;
end bench;
```

DisplayDecoder_constr.xdc

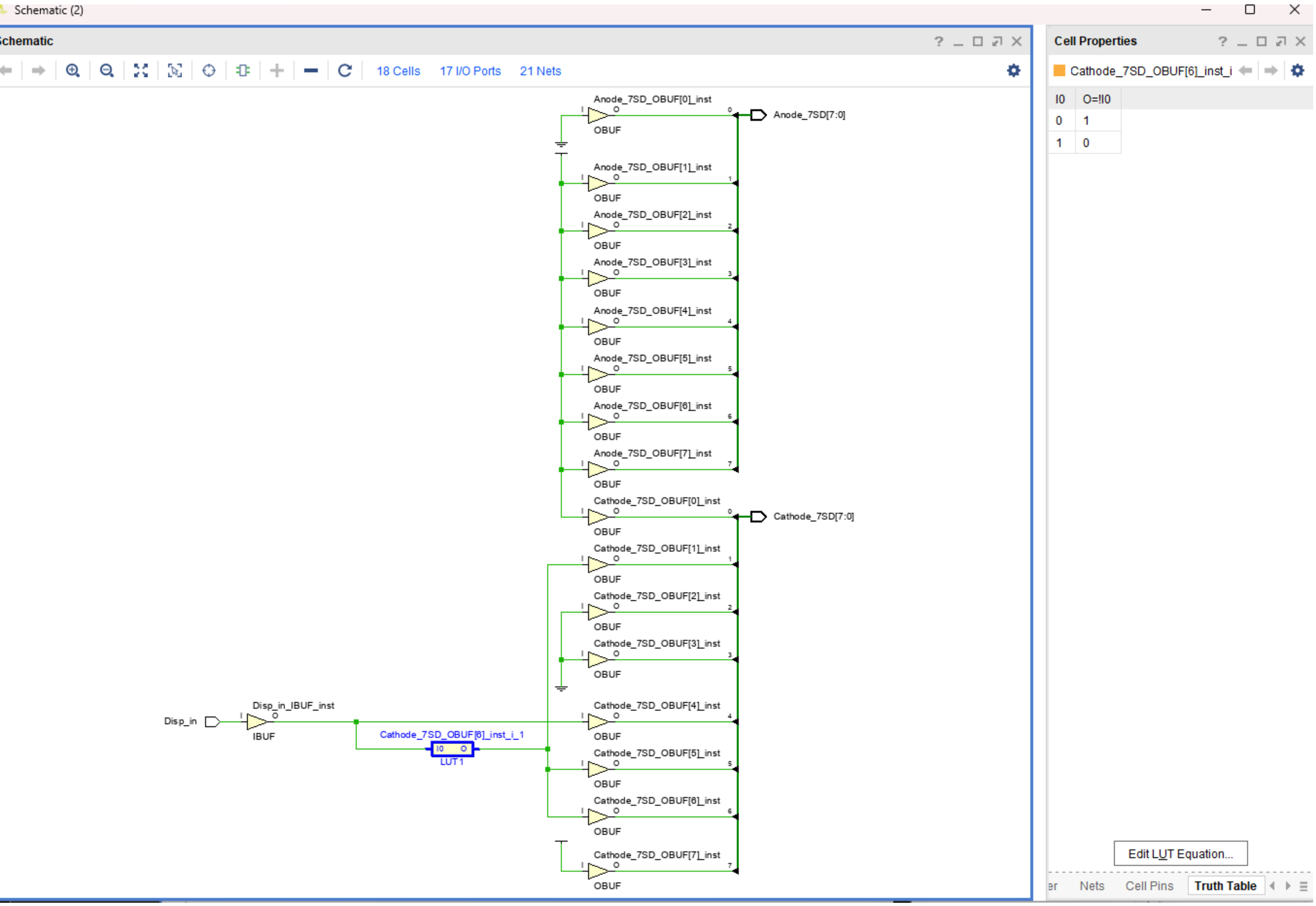
:25/ECE238L/DisplayDecoder/DisplayDecoder.srcs/constrs_1/new/DisplayDecoder_constr.xd

```
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[7]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[6]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[5]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[4]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[7]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[6]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[5]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[4]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports Disp_in]
set_property PACKAGE_PIN J15 [get_ports Disp_in]
set_property PACKAGE_PIN T10 [get_ports {Cathode_7SD[7]}]
set_property PACKAGE_PIN R10 [get_ports {Cathode_7SD[6]}]
set_property PACKAGE_PIN K16 [get_ports {Cathode_7SD[5]}]
set_property PACKAGE_PIN K13 [get_ports {Cathode_7SD[4]}]
set_property PACKAGE_PIN P15 [get_ports {Cathode_7SD[3]}]
set_property PACKAGE_PIN T11 [get_ports {Cathode_7SD[2]}]
set_property PACKAGE_PIN L18 [get_ports {Cathode_7SD[1]}]
set_property PACKAGE_PIN H15 [get_ports {Cathode_7SD[0]}]
set_property PACKAGE_PIN U13 [get_ports {Anode_7SD[7]}]
set_property PACKAGE_PIN K2 [get_ports {Anode_7SD[6]}]
set_property PACKAGE_PIN T14 [get_ports {Anode_7SD[5]}]
set_property PACKAGE_PIN P14 [get_ports {Anode_7SD[4]}]
set_property PACKAGE_PIN J14 [get_ports {Anode_7SD[3]}]
set_property PACKAGE_PIN T9 [get_ports {Anode_7SD[2]}]
set_property PACKAGE_PIN J18 [get_ports {Anode_7SD[1]}]
set_property PACKAGE_PIN J17 [get_ports {Anode_7SD[0]}]
```

# Display Decoder Waveform



# Display Decoder Truth Table and Schematic




| I0 | O=!I0 |
|---|---|
| 0 | 1 |
| 1 | 0 |

# Top Module VHDL Source Code, Simulation Code and Constraints

Lab3.vhd

C:/Users/tsalaz25/ECE238L/Lab3/Lab3.srcs/sources_1/new/Lab3.vhd

```vhdl
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

-- Uncomment the following library declaration if using
-- arithmetic functions with Signed or Unsigned values
--use IEEE.NUMERIC_STD.ALL;

-- Uncomment the following library declaration if instantiating
-- any Xilinx leaf cells in this code.
--library UNISIM;
--use UNISIM.VComponents.all;

entity Lab3 is
  Port ( A, B : in STD_LOGIC;
         Selector : in STD_LOGIC_VECTOR (2 downto 0);
         Mux_out : out STD_LOGIC;
         Cathode_7SD : out STD_LOGIC_VECTOR (7 downto 0);
         Anode_7SD : out STD_LOGIC_VECTOR (7 downto 0));
end Lab3;

architecture Behavioral of Lab3 is
-- Component from Gates
component Gates
  Port ( A, B : in STD_LOGIC;
         G_out : out STD_LOGIC_VECTOR (7 downto 0));
end component;
signal G_out : STD_LOGIC_VECTOR (7 downto 0);
-- Component from Multiplexer
component Multiplexer
  Port ( Mux_in : in STD_LOGIC_VECTOR (7 downto 0);
         Selector : in STD_LOGIC_VECTOR (2 downto 0);
         Mux_out : out STD_LOGIC);
end component;
signal Mux_in : STD_LOGIC_VECTOR (7 downto 0);
-- Component from Display Decoder
component DisplayDecoder
  Port ( Disp_in : in STD_LOGIC;
         Cathode_7SD : out STD_LOGIC_VECTOR (7 downto 0);
         Anode_7SD : out STD_LOGIC_VECTOR (7 downto 0));
end component;
signal Mux : STD_LOGIC;

begin
  lg : Gates port map (A => A, B => B, G_out => G_out);
  muxx : Multiplexer port map (Selector => Selector,
                               Mux_in => G_out,
                               Mux_out => Mux);
  disp : DisplayDecoder port map (Disp_in => Mux,
                                  Cathode_7SD => Cathode_7SD,
                                  Anode_7SD => Anode_7SD);
  Mux_out <= Mux;
end Behavioral;
```

Lab3_tb.vhd

C:/Users/tsalaz25/ECE238L/Lab3/Lab3.srcs/sim_1/new/Lab3_tb.vhd

```vhdl
entity Lab3_tb is
--  Port ( );
end Lab3_tb;

architecture bench of Lab3_tb is
component Lab3
  Port ( A, B     : in STD_LOGIC;
         Selector : in STD_LOGIC_VECTOR (2 downto 0);
         Mux_out  : out STD_LOGIC;
         Cathode_7SD : out STD_LOGIC_VECTOR (7 downto 0);
         Anode_7SD   : out STD_LOGIC_VECTOR (7 downto 0));
end component;

signal A_tb, B_tb : STD_LOGIC;
signal Selector_tb : STD_LOGIC_VECTOR(2 downto 0);
signal Mux_out_tb : STD_LOGIC;
signal Cathode_7SD_tb : STD_LOGIC_VECTOR (7 downto 0);
signal Anode_7SD_tb : STD_LOGIC_VECTOR (7 downto 0);

begin
  uut : Lab3 port map ( A => A_tb,
                        B => B_tb,
                        Mux_out => Mux_out_tb,
                        Selector => Selector_tb,
                        Cathode_7SD => Cathode_7SD_tb,
                        Anode_7SD => Anode_7SD_tb);

stimulus : process
  begin
    Selector_tb <= "000" ; A_tb <= '0' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "001" ; A_tb <= '0' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "010" ; A_tb <= '0' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "011" ; A_tb <= '0' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "100" ; A_tb <= '0' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "101" ; A_tb <= '0' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "110" ; A_tb <= '0' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "111" ; A_tb <= '0' ; B_tb <= '0' ; wait for 100 ns;

    Selector_tb <= "000" ; A_tb <= '0' ; B_tb <= '1' ; wait for 100 ns;
    Selector_tb <= "001" ; A_tb <= '0' ; B_tb <= '1' ; wait for 100 ns;
    Selector_tb <= "010" ; A_tb <= '0' ; B_tb <= '1' ; wait for 100 ns;
    Selector_tb <= "011" ; A_tb <= '0' ; B_tb <= '1' ; wait for 100 ns;
    Selector_tb <= "100" ; A_tb <= '0' ; B_tb <= '1' ; wait for 100 ns;
    Selector_tb <= "101" ; A_tb <= '0' ; B_tb <= '1' ; wait for 100 ns;
    Selector_tb <= "110" ; A_tb <= '0' ; B_tb <= '1' ; wait for 100 ns;
    Selector_tb <= "111" ; A_tb <= '0' ; B_tb <= '1' ; wait for 100 ns;

    Selector_tb <= "000" ; A_tb <= '1' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "001" ; A_tb <= '1' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "010" ; A_tb <= '1' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "011" ; A_tb <= '1' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "100" ; A_tb <= '1' ; B_tb <= '0' ; wait for 100 ns;
```

Lab3_constr.xdc

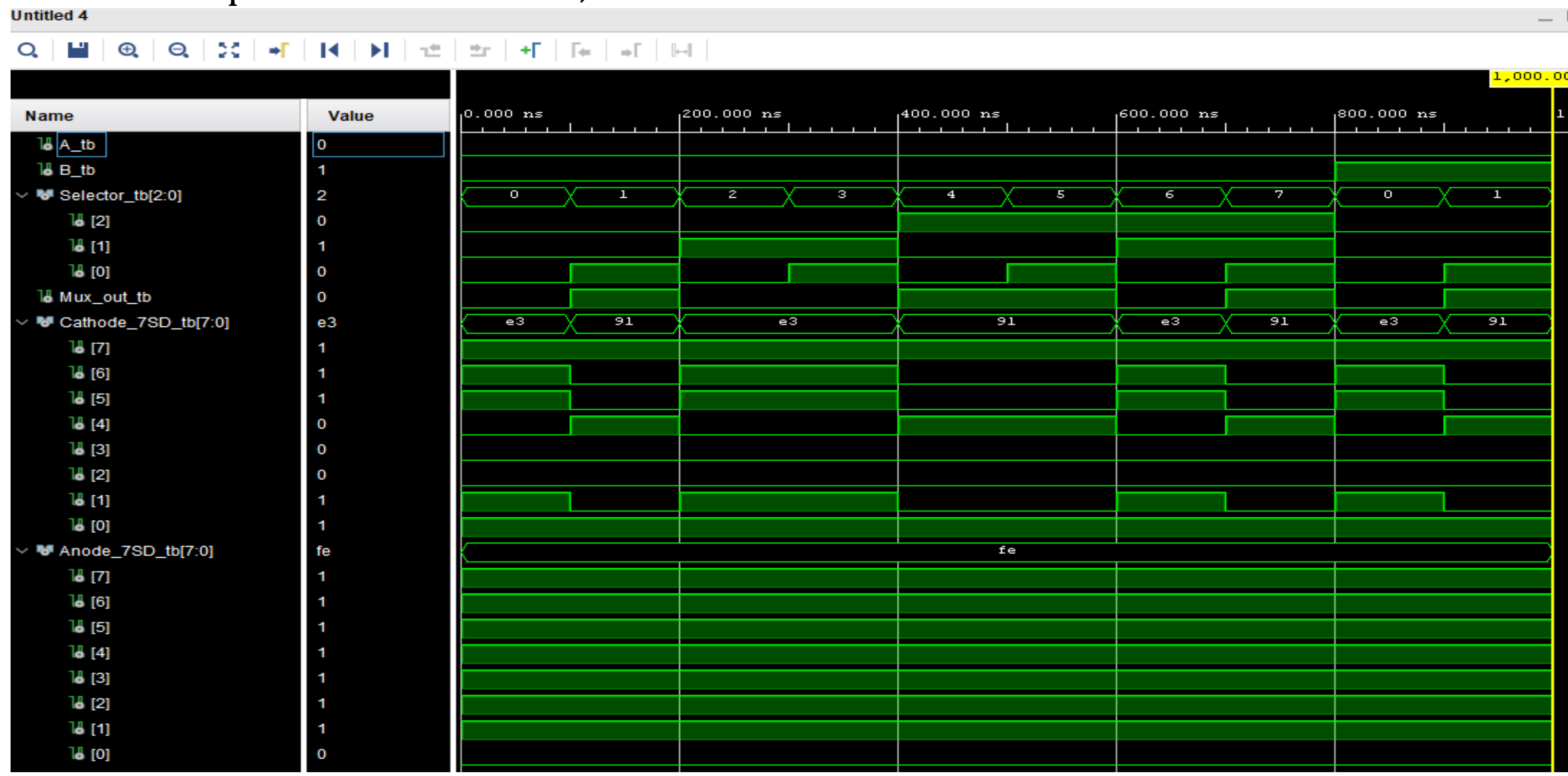
C:/Users/tsalaz25/ECE238L/Lab3/Lab3.srcs/constrs_1/new/Lab3_constr.xdc

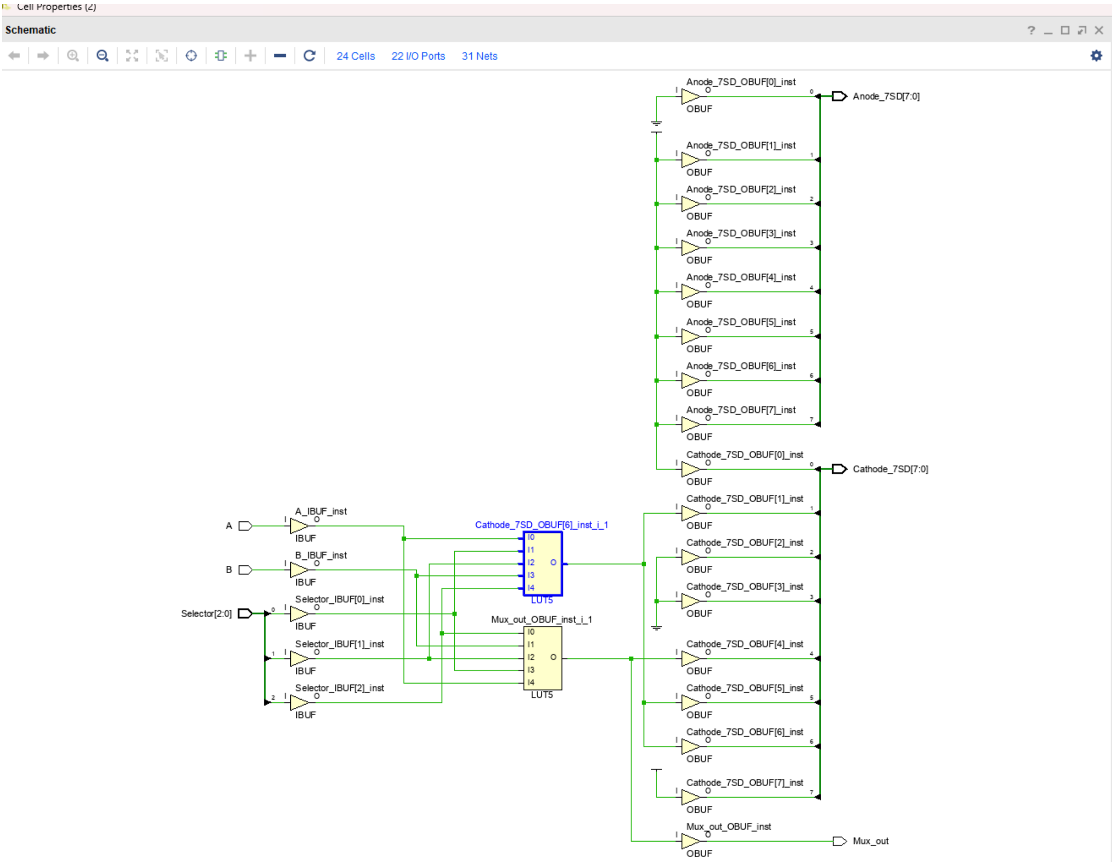
This file has been changed. Reload

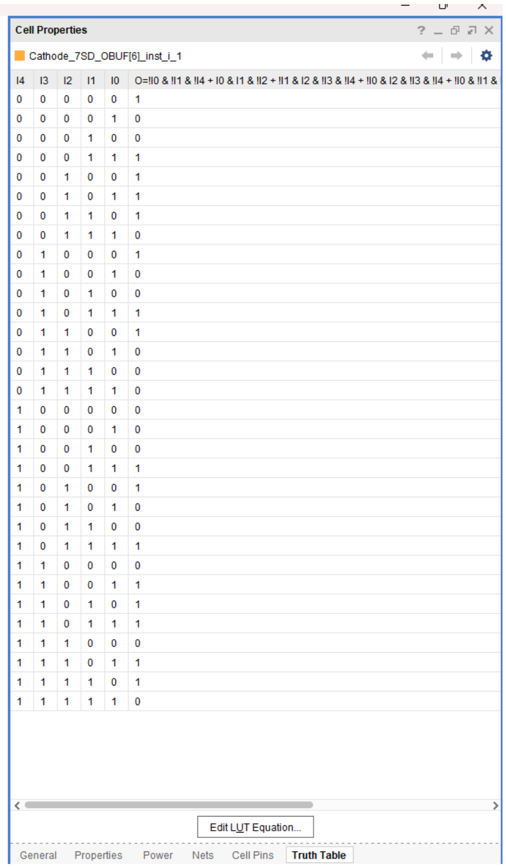
```
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[7]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[6]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[5]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[4]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Anode_7SD[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[7]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[6]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[5]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[4]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[3]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Cathode_7SD[0]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Selector[2]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Selector[1]}]
set_property IOSTANDARD LVCMOS33 [get_ports {Selector[0]}]
set_property PACKAGE_PIN J15 [get_ports A]
set_property PACKAGE_PIN L16 [get_ports B]
set_property PACKAGE_PIN K15 [get_ports Mux_out]
set_property PACKAGE_PIN U12 [get_ports {Selector[0]}]
set_property PACKAGE_PIN U11 [get_ports {Selector[1]}]
set_property PACKAGE_PIN V10 [get_ports {Selector[2]}]
set_property PACKAGE_PIN H15 [get_ports {Cathode_7SD[0]}]
set_property PACKAGE_PIN L18 [get_ports {Cathode_7SD[1]}]
set_property PACKAGE_PIN T11 [get_ports {Cathode_7SD[2]}]
set_property PACKAGE_PIN P15 [get_ports {Cathode_7SD[3]}]
set_property PACKAGE_PIN K13 [get_ports {Cathode_7SD[4]}]
set_property PACKAGE_PIN K16 [get_ports {Cathode_7SD[5]}]
set_property PACKAGE_PIN R10 [get_ports {Cathode_7SD[6]}]
set_property PACKAGE_PIN T10 [get_ports {Cathode_7SD[7]}]
set_property PACKAGE_PIN J17 [get_ports {Anode_7SD[0]}]
set_property PACKAGE_PIN J18 [get_ports {Anode_7SD[1]}]
set_property PACKAGE_PIN T9 [get_ports {Anode_7SD[2]}]
set_property PACKAGE_PIN J14 [get_ports {Anode_7SD[3]}]
set_property PACKAGE_PIN P14 [get_ports {Anode_7SD[4]}]
set_property PACKAGE_PIN T14 [get_ports {Anode_7SD[5]}]
set_property PACKAGE_PIN K2 [get_ports {Anode_7SD[6]}]
set_property PACKAGE_PIN U13 [get_ports {Anode_7SD[7]}]
```

```vhdl
    Selector_tb <= "000" ; A_tb <= '1' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "001" ; A_tb <= '1' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "010" ; A_tb <= '1' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "011" ; A_tb <= '1' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "100" ; A_tb <= '1' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "101" ; A_tb <= '1' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "110" ; A_tb <= '1' ; B_tb <= '0' ; wait for 100 ns;
    Selector_tb <= "111" ; A_tb <= '1' ; B_tb <= '0' ; wait for 100 ns;

    Selector_tb <= "000" ; A_tb <= '1' ; B_tb <= '1' ; wait for 100 ns;
    Selector_tb <= "001" ; A_tb <= '1' ; B_tb <= '1' ; wait for 100 ns;
    Selector_tb <= "010" ; A_tb <= '1' ; B_tb <= '1' ; wait for 100 ns;
    Selector_tb <= "011" ; A_tb <= '1' ; B_tb <= '1' ; wait for 100 ns;
    Selector_tb <= "100" ; A_tb <= '1' ; B_tb <= '1' ; wait for 100 ns;
    Selector_tb <= "101" ; A_tb <= '1' ; B_tb <= '1' ; wait for 100 ns;
    Selector_tb <= "110" ; A_tb <= '1' ; B_tb <= '1' ; wait for 100 ns;
    Selector_tb <= "111" ; A_tb <= '1' ; B_tb <= '1' ; wait for 100 ns;



  wait;
  end process;
end bench;
```

# Top Module Waveform, Truth Table and Schematic



Cell Properties

Cathode_7SD_OBUF[6]_inst_i_1

| I4 | I3 | I2 | I1 | I0 | O=!I0 & !I1 & !I4 + I0 & I1 & !I2 + !I1 & I2 & !I3 & !I4 + !I0 & I2 & !I3 & !I4 + !I0 & !I1 & |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 0 | 1 | 1 | 1 |
| 0 | 0 | 1 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 1 | 1 |
| 0 | 0 | 1 | 1 | 0 | 1 |
| 0 | 0 | 1 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 | 0 |
| 0 | 1 | 0 | 1 | 1 | 1 |
| 0 | 1 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 0 |
| 1 | 0 | 0 | 1 | 0 | 0 |
| 1 | 0 | 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 | 1 | 0 |
| 1 | 0 | 1 | 1 | 0 | 0 |
| 1 | 0 | 1 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 | 0 |

Edit LUT Equation...

General | Properties | Power | Nets | Cell Pins | Truth Table

Summary:
Lab 3 was like Lab 2 in the sense that we made multiple parts that were all implemented in a Top Module. I liked this Lab, but it was a bit tedious. Gates as used to implement Logic Gates (AND, OR, NOT, NOR, NAND, XOR and XNOR). The Multiplexer was next, out Inputs for the MUX came from the output of Gate, and Multiplexer output a single bit to represent a high or low signal. The last component was the Display Decoder. This was used to display the signal from the MUX as either H (high) or L (low). The Top Modules is where the entire project comes together. In this module we implemented the Gates, MUX and Display to make the ATRIX-7 machine functional for the desired outputs in the lab, which was dependent on the inputs for A and B and the input for the mux selector.

Problems:
There weren't any real problems, I think understanding the components and HOW they connect was one of my issues. But as the project came together it made more sense. So no real problems with VHDL, I think the syntax is becoming more normal at this point. The lab structure was my biggest problem.

Hints:
Make sure you know what is going on, there are a lot more variables in this lab, which can make it harder to keep track. In previous labs you could get away with naming the variables the same thing, not as easy in this lab.

Recommendations:
It's a good lab, I think the showing us the end goal more explicitly can help, I know we have the board demo video but in the moment, that made me more confused.